# Tanisha Krishnaraj

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## Education

**Boston University** Boston, MA

*Bachelor of Science in Applied Mathematics & Data Science (GPA: 3.84)* *Expected May 2028*

- **Honors/Awards:** Dean's List (All Semesters)
- **Relevant Coursework:** Computational Linear Algebra, Probability Theory, Stochastic Processes, Algorithms

## Technical Skills

**Languages:** Julia, Rust, C, Python (NumPy, Pandas, Scikit-learn, PyTorch), R, SQL
**Mathematical Modeling:** Numerical Optimization, Spectral Methods, Stochastic Modeling
**Systems & Infrastructure:** Linux (Bash), OpenSSL, TCP/IP, Encryption, EDR (CrowdStrike Falcon, Carbon Black)

## Research Experience

**Directed Reading Program (DRP) | Boston University** Boston, MA

*Research Fellow — Algebraic Graph Theory & Spectral Methods* *February 2026 – May 2026*

- Studied network modeling through graph-theoretic representations under PhD supervision, investigating the spectral properties and eigenvalue distributions of adjacency and Laplacian matrices.
- Explored Godsil and Royle's Algebraic Graph Theory, including graph coloring, combinatorial counting, and structural invariants; met regularly with a mentor to present findings and incorporate feedback.

## Technical Projects

**Loss Landscape & Gradient Descent Analysis** | *Python, NumPy, Matplotlib* October 2026

- Implemented and analyzed gradient descent for housing price regression, reducing Hessian condition number from $5.88 \times 10^{10}$ to 20.88 through feature scaling.
- Derived optimal step size analytically, accelerating gradient descent convergence by 28× (from 5,000 down to 176 iterations) and matching closed-form regression coefficients within $3.84 \times 10^{-6}$.

## Work Experience

**Information Security Compliance | Boston University** Boston, MA

*Security Assistant* *March 2026 – Present*

- Managed endpoint security configuration and system integrity for 200+ research nodes using CrowdStrike Falcon and VMware Carbon Black to prevent unauthorized data exfiltration.
- Engineered automated Bash and Python routines to parse vulnerability reports, streamline ServiceNow tracking, and deploy systematic configuration patches.

**Center for Computing & Data Sciences | Boston University** Boston, MA

*Course Facilitator: Linear Algebra* *January 2026 – May 2026*

- Mentored 60+ students on Singular Value Decomposition, Spectral Decomposition, Vector Spaces, and Markov Chains, explaining matrix transformations clearly in writing and in person.
- Organized weekly technical meetings to bridge abstract theory with practical algorithmic implementations of linear algebra.

## Leadership Experience

**PROMYS Pathways** Boston, MA

*Mentor* *June 2026 – August 2026*

- Mentored a 30-student cohort through intensive number theory and modular arithmetic concepts, using inquiry-based guidance to build mathematical intuition over rote instruction.
- Facilitated weekly collaborative problem-solving sessions, coaching students to discover proofs independently and self-correct their reasoning.